4 Which of the following statements is true?

A When a moving object experiences friction, the work done against friction is independent of the path taken by the object.

B An object which experiences a resistive force always reaches terminal speed.

C Friction always hinders motion.

D There are certain situations in which friction does not cause energy to be lost to the surroundings.

Ans: D

For example, if you push an object and the friction acting on it equals your pushing force, the object will not move. Then the work done against friction, which is the energy lost to the surroundings, will be zero.

Option A: the work done against friction is path dependent.
Option B: a falling object will reach terminal speed only if there is enough distance for it to fall before hitting the ground.
Option C: Friction between our feet and the ground is what allows us to move forward.